

EXHIBIT 2



Refractory Metal

In subject area: [Materials Science](#)

Refractory metals are defined as selected metals with melting points around 2000 °C and higher, including Tungsten, Molybdenum, Chromium, Tantalum, Niobium, and Rhenium, along with their refractory alloys.

AI generated definition based on: [International Journal of Refractory Metals and Hard Materials, 2015](#)

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Chapter

Refractory Alloys: Vanadium, Niobium, Molybdenum, Tungsten

2019, *Structural Alloys for Nuclear Energy Applications*

Lance L. Snead, ... Andre A.N. Nemith

Abstract

Refractory metals as a class of materials are understood to share the common properties of very high melting temperature and mechanical properties and wear resistance. A narrowly defined class of refractory metals would include metals with melting points >2000°C: [niobium](#), chromium, molybdenum, [tantalum](#), [tungsten](#), and [rhenium](#) [1], while a wider class would also include those with melting points above 1850°C: vanadium, [hafnium](#), titanium, [zirconium](#), [ruthenium](#), [osmium](#), [rhodium](#), and [iridium](#). The current practical application of refractory metals is relatively widespread (though arguably for specialty application) with examples being casting molds, wire filaments, reactant vessels for corrosive materials, hard tooling, and a myriad of applications where high density is desired. Because refractory metals are a class of materials possessing extraordinary high-temperature properties, they are perennial contenders for high-temperature nuclear applications. However, their use to date has been limited, due in part to the difficulty in fabricating high-performance refractory parts and their environmental degradation including irradiation effects. The following sections will discuss the current processing routes being taken to produce nuclear-grade [refractory alloys](#), a general discussion of their properties, and the effects of irradiation on the materials.

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Material evaluation and process selection

2003, [Process Planning](#)

Peter Scallan

Refractory metals

The most common metals in this group are molybdenum, [niobium](#), [tantalum](#) and [tungsten](#). The major use of these metals is as alloying elements for steel. Finally, they typically have melting points above 1600 °C and thus the name [refractory metals](#). Even at such high temperatures, they maintain their [strength](#).

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Chapter

Refractory Metals and Alloys

2016, [Reference Module in Materials Science and Materials Engineering](#)

C.L. Briant, M.K. Banerjee

5 Concluding Remarks

This article reviews only the basic physical properties of the four [refractory metals](#), i.e., preparation, processing, and common alloys based on these materials. Refractory metals still remain a special class of materials because of their combination of high melting points and extreme reactivity with oxygen at elevated temperatures. Today, they are primarily used as alloying elements in steels and super alloys, or for niche markets.

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Review article

Refractory metals revolutionizing the lighting technology: A historical review

2015, [International Journal of Refractory Metals and Hard Materials](#)

P. Schade, ... I. Smid

Preamble

[Refractory metals](#)

The Americans call them “exotic”

The British “less common”

The European Union “sensitive”

We stay with the truth and call them “refractory”.

Walter M. Schwarzkopf, 1972

“The refractory metals comprise selected metals with melting points around 2000°C and higher, with similar powder metallurgical production and application. This includes the following metals and their [refractory alloys](#): [Tungsten](#), Molybdenum, Chromium, [Tantalum](#), [Niobium](#) and Rhenium” [1].

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Environmental degradation of structural materials in liquid lead- and lead-bismuth eutectic-cooled reactors

2022, *Progress in Materials Science*

Xing Gong, ... Konstantina Lambrinou

2.3.2 Refractory metals

Refractory metals, such as W, Mo, Nb, and Ta, are attractive materials for engineering applications at elevated temperatures, due to their high melting points and excellent strength. These materials have good creep resistance, void swelling resistance, and very low solubility in oxygen-poor liquid Pb and LBE. In this regard, they show potential for future use in LFRs. However, their application in a nuclear system is challenged by their strong susceptibility to low-temperature irradiation embrittlement, weak oxidation resistance (potentially problematic for use in HLMs with high oxygen contents), bad weldability, and manufacturing difficulty [64].

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Chapter

Refractory Alloys: Vanadium, Niobium, Molybdenum, Tungsten

2019, *Structural Alloys for Nuclear Energy Applications*

Lance L. Snead, ... Andre A.N. Nemith

13.4.5 Summary and Conclusions

Refractory metals have historically been, and continue to be, perennial candidates for high-temperature nuclear systems. However, their application has been predominantly notional or limited to nonstructural applications due largely to the issues highlighted in this chapter: challenging manufacturing issues associated with sensitivity to O, C, N pickup, a structure with inherently low as-fabricated ductility in most cases (particularly for Mo and W alloys), and pronounced embrittlement due to neutron irradiation. While the irradiation-induced embrittlement at the crystal level is extremely challenging to mitigate for this class of materials, strides are being made in our ability to both fabricate complicated (e.g., graded composition layers) refractory structures and in our understanding of how to design structures with low-ductility materials. Given this coupled processing and design understanding, it is becoming more likely that refractory materials will find more widespread and important application within nuclear systems in the near future.

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Chapter

Refractory Metals and Alloys

2016, *Reference Module in Materials Science and Materials Engineering*

C.L. Briant, M.K. Banerjee

Abstract

This module introduces five important **refractory metals**. The physical properties along with the common characteristics of the concerned refractory metals are described. The manufacturing methods of the individual refractory metals and the constraints thereof are enumerated. The properties and uses of refractory metals under consideration are also discussed. Following this, various alloys of these refractory metals, their novel properties, and specific uses are elaborated. Pseudo softening and **dynamic strain ageing** in some dilute **binary alloys** are also recorded. A comparative analysis of **strength** and weakness of different alloys are presented.

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Review article

Additive manufacturing of metallic components – Process, structure and properties

2018, *Progress in Materials Science*

T. DebRoy, ... W. Zhang

5.1 Refractory alloys

Refractory metal alloys such as **tungsten**, **tantalum**, **niobium**, molybdenum, vanadium, chromium and **rhenium** are the highest melting point metals and have BCC or HCP crystal structures. These metals and alloys have unique properties and applications that include high temperature **strength**, biocompatibility, low thermal expansion, **superconductivity**, high density and used in structural applications in high radiation environments. Refractory metals are reactive at elevated temperatures and can lose ductility when processed with insufficient shielding from atmospheric contamination. These alloys are often welded in vacuum using the PBF-EB. Development efforts for building of complex structures with these alloys suitable for use in extreme environments [574] in the aerospace and nuclear industries dates back to the 1960s. Limitations in applying these materials more widely include their high cost, a limited supply of commercial shapes, and difficulties in fabrication using conventional metal process technology. As a result, **AM** is a candidate technology to expand the use of these materials as it can reduce the need for conventional processing steps, produce commercial shapes and use significantly smaller quantities of expensive materials. Early work in the laser powder fusing of metals into complex shapes using computer models and control demonstrated the potential of the technology to refractory metal components [575,576]. Because of this work, subsequent technology advances focused on the development of high purity of spherical metal powders better suited for additive applications [577].

Tungsten alloys provide low thermal expansion and high strength at high temperatures and have found applications in furnace applications and components experiencing high heat flux. Due to its high density it has found application in radiation shielding, energy beam collimation and balance weight applications. While conventional processing

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Chapter

Recycling of Rare Metals

2014, *Handbook of Recycling*

10.4 Refractory Metals (Ferro-alloys Metals, Specialty Metals)

Refractory metals (RMs) are high melting point metals that are characterized by other special physical and chemical properties, such as high density, inertness, corrosion and acid resistance. They are produced both as metal ingot (buttons) using electron beam furnaces and as metal powder that serves as raw material for powder metallurgical treatments like pressing and sintering. The definition of which elements belong to this group differs. As defined at the EU level (European IPPC Bureau, 2013), this group comprises 11 metals, the elements of the fourth to the seventh transition group of the periodic table. Due to their main applications (see U.S. Geological Survey, 2013; European IPPC Bureau, 2013; European Commission-Enterprise and Industry, 2010) they can be subdivided as:

- *Ferroalloy metals (RMs for steel production)*: chromium (Cr), manganese (Mn), molybdenum (Mo), niobium (Nb), vanadium (V), and
- *Specialty metals (RMs for special applications)*: hafnium (Hf), tantalum (Ta), titanium (Ti), rheneum (Re), tungsten (W), and zirconium (Zr).

Ferro-alloys: Ferro-alloys are mainly used as master alloys in the iron, foundry and steel industry, because it is the most economical way to introduce an alloying element into the steel melt. Besides this, special ferroalloys are also needed for the production of aluminum alloys and as a starting material in specific chemical reactions. As an additive in steel production, ferroalloys improve steel's properties, especially tensile strength, wear and corrosion resistance. In 2012, more than 90% of the produced Cr, Mn, Mo and V (mine productions: 24Mt Cr, 16Mt Mn, 250,000t Mo, 73,000t V) as well as about 80% of the produced Nb (mine production: 69,000t) were used in this sector.

Superalloys and special alloys: In 2012 approximately 70% of Re (mine production: 52t)

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Chapter

Effect of Radiation on Strength and Ductility of Metals and Alloys

2012, *Comprehensive Nuclear Materials* (Second Edition)

M.L. Grossbeck

1.05.8 Refractory Metals

1.05.8.1 Tensile Behavior

The refractory metals are the metals in groups V and VI of the periodic table: vanadium, niobium, and tantalum in group V and chromium, molybdenum, and tungsten in group VI. All have the characteristic of a high melting point, hence the term refractory. The group VI metals are typically brittle, even without irradiation. For example, chromium is almost never used pure or as a major alloy element, although it is invaluable as a minor alloying element. Molybdenum and tungsten are both brittle in nature but can be made into useful structural alloys by controlling interstitial impurities and by the addition of minor elements. In contrast to the brittle behavior of the group VI metals, the group V metals are inherently ductile. Structural alloys based upon this group have been developed, primarily for very high temperature and space applications. The primary disadvantage of the refractory metals is their formation of volatile oxides as opposed to protective oxide layers. Vanadium and molybdenum oxides have melting points below metal working temperatures so that the metals become wet and can have

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liquid oxide drip off them.

Unlike the [tensile behavior of fcc metals](#), where there is a smooth increase in [strength](#) as [plastic deformation](#) proceeds and work hardening progresses, [bcc metals](#) typically exhibit a load drop, or yield point, almost immediately following the onset of plastic deformation, as discussed in Section 1.05.7.2.

In the case of refractory metals, mechanical properties are largely determined by interstitial solutes. High purity refractory metals do not exhibit a yield point but behave more like [fcc metals](#). [Niobium alloys](#) irradiated in Li at 1200°C for over three months in EBR-II had total elongations of about 60%. Despite any irradiation hardening, the near

absence of oxygen resulted in a very soft material at these high temperatures. 50

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